

Team: On Jeti

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Course: SDS 106 - Algorithms and Programming 2 / Educational Practice

Project Report

Introduction

Our team has chosen www.shanghairanking.com for a Data Scraping, Visualization and Analysis project. This is a website from the educational domain. It contains different university rankings. Our main objective was to identify the main factors that ascend university rank in global rankings. Since we are primarily interested in Data Science and adjacent fields, we mainly focused on historical data in Mathematics and Statistics ranking.

Motivation

Beyond academic requirements, this study is driven by a desire to identify the institutional qualities that foster world-class research and learning environments. As we prepare for post-graduate studies, understanding these metrics helps us evaluate potential academic destinations. Furthermore, this analysis aims to provide actionable insights for the development of the higher education sector in **Kazakhstan**, identifying how local institutions can strategically improve their global standing.

Data Source Description

We extracted two primary datasets for cross-analysis:

1. **Subject-Specific Historical Data:** We scraped all available records for Mathematics and Statistics. The data structure is longitudinal, identified by the URL suffix (e.g., /2025/AS0101), where the year and a six-character subject code allow for time-series analysis.
2. **Institutional Metadata:** A second dataset containing up-to-date global rankings and university foundational data was scraped to test the correlation between **institutional age** and current research quality.

Technical Execution: The website architecture employs a mix of static and dynamic (JavaScript-rendered) elements. To navigate these complexities, we implemented a hybrid scraping pipeline: **Selenium** was utilized to handle pagination and asynchronous content loading, while **BeautifulSoup** was used for high-efficiency data extraction from the rendered HTML.

Ethics: No robots.txt exists for this site, but we used request delays not to overwhelm the server.

Scraping Methodology

Our data collection process involved scraping two distinct tables, each requiring a specialized approach due to varying website architectures.

1. Primary Ranking Table (7,000+ Rows, 9 Columns)

The first table was systematically extracted by generating unique URLs for every combination of academic year and subject.

- ❖ **Navigation:** This table was very clearly constructed and the only complexity we faced was the pagination system. The website link did not change dynamically after clicking to the next page, so we decided to use Selenium to mimic human activity.
- ❖ **Overcoming Detection Issue:** We set up a User-Agent header for Selenium via [options](#). This header is one that appears on my (Zhanar) laptop if I surf this website, so it is real-world based.
- ❖ **Execution:** The script navigated to each URL via `driver.get()`, captured the HTML source (`driver.page_source`), and extracted the data. We used the **Expected Conditions** module to ensure the DOM had fully loaded after interacting with the button to the next page.

2. Supplementary University Profile Data (3,000+ Rows, 17 Columns)

The second table was really troublesome and we decided to fully stick to Selenium.

- ❖ **Navigation Strategy:** This table contains a list of ranked universities and each uni name navigates us to a page with full info about the university. We were particularly interested in the founded year column. Buttons on this site do not contain any href links so we were ought to manually click to each element and use `driver.back()` command to return to the main page.
- ❖ **Data Extraction:** Information was gathered via CSS selectors.
- ❖ **State Management:** To handle potential crashes during this intensive process, we built a **recovery mechanism**. If the script failed, it would check the current page index and use a «skip 5 pages» function to quickly navigate back to the last successful state.
- ❖ **Persistence:** Data was saved to a CSV file every 30 records to ensure progress was not lost during long-running sessions.

3. Reliability and Ethical Scraping

For both datasets, we prioritized script stability and ethical crawling:

- ❖ **Error Handling:** We utilized try-except-finally blocks for comprehensive logging and error diagnosis.
- ❖ **Human Mimicry:** To prevent server strain, we implemented a **randomized request delay** using `time.sleep(np.random.uniform(1, 2))`.
- ❖ **Data Availability:** The final raw datasets—totaling over 10,000 rows—have been hosted on Kaggle for accessibility.

4. Kaggle dataset links

- ❖ [The First Dataset](#)
- ❖ [The Second Dataset](#)

The thing with scraping, that the first dataset was ready in mere 5 minutes and the second one took as much as 6 hours.

Cleaning Steps

1. Handling Missing Values

- ❖ **Action:** Imputation via logical summation.
- ❖ **Logic:** We observed missing values in the Total Score column. Since the total score is a composite metric of specific performance indicators, we imputed missing values using the sum of: *World-Class Output + World-Class Faculty + High Quality Research + Research Impact + International Collaboration*.
- ❖ **Justification:** This approach is more accurate than mean or median imputation because it reconstructs the missing value based on the university's actual performance in individual sub-categories, maintaining the internal consistency of the record.

2. Deduplication

- ❖ **Action:** Identification and removal of exact duplicates.
- ❖ **Logic:** Using `df.drop_duplicates()`, we filtered out records where every column was identical.
- ❖ **Justification:** Duplicates often occur in web scraping due to pagination overlaps or network retries. Removing them prevents the artificial inflation of university performance metrics and ensures each data point is unique.

3. Text Normalization & Structural Extraction

- ❖ **Action:** Parsing the Institution and Year/Subject columns.
- ❖ **Normalization:** We used `.str.strip()` to remove leading/trailing whitespace and `.str.split("\n")` to separate the university name from its geographic region.
- ❖ **Mapping:** The Year/Subject column contained concatenated strings (e.g. «2021-AS0101»). We extracted the first four characters as the Year and mapped internal codes (like «AS0101») to human-readable names like **Mathematics** and **Statistics**.
- ❖ **Justification:** This separates distinct variables stored in a single string, enabling categorical filtering by Subject and Region.

4. Standardization of Data Types

- ❖ **Action:** Temporal and Numerical conversion.
- ❖ **DateTime:** Converted the extracted Year string into a datetime object and then back to an integer representing the year.
- ❖ **Numeric:** Ensured all score-related columns were formatted as floats.
- ❖ **Justification:** Standardizing data types allows for time-series analysis and mathematical operations (like calculating the mean score over time) that are impossible on «object» (string) types.

5. Outlier Detection

- ❖ **Action:** Z-score analysis and Boxplot visualization.
- ❖ **Logic:** We calculated **Z-scores** for all numeric columns. A threshold of $|Z| > 3$ was used to identify statistical outliers.
- ❖ **Justification:** We chose **not to blindly remove these outliers**. In university rankings, «Elite» universities often perform significantly better than the global average. Removing them would erase the most successful institutions in the dataset. Instead, we visualize them using boxplots to understand the spread of prestige.

6. Feature Engineering

We derived three new columns to provide deeper insights into the dataset:

- ❖ **Research Impact Factor:** Calculated as Research Impact / High Quality Research.
 - *Rationale:* This measures efficiency—how much impact is generated per unit of high-quality research output.
- ❖ **Total Score Category:** Using `pd.cut`, we binned scores into **Standard**, **Superior**, and **Elite**.

- **Rationale:** This simplifies complex decimals into a categorical ranking for easier comparison in visualizations.
- ❖ **Continent Mapping:** We mapped the Region (Country) to a broader **Continent** category.
 - **Rationale:** This allows for macro-level analysis of academic performance by global region (e.g., comparing European vs. Asian academic growth).

7. Before vs After

Data columns (total 9 columns):

#	Column	Non-Null Count	Dtype
0	World Rank	7104 non-null	object
1	Institution	7104 non-null	object
2	Total Score	1003 non-null	float64
3	World-Class Faculty	7104 non-null	float64
4	World-Class Output	7104 non-null	float64
5	High Quality Research	7104 non-null	float64
6	Research Impact	7104 non-null	float64
7	International Collaboration	7104 non-null	float64
8	Year/Subject	7104 non-null	object

Before

#	Column	Non-Null Count	Dtype
0	World Rank	6402 non-null	object
1	Region	6402 non-null	object
2	University	6402 non-null	object
3	Total Score	6402 non-null	float64
4	World-Class Faculty	6402 non-null	float64
5	World-Class Output	6402 non-null	float64
6	High Quality Research	6402 non-null	float64
7	Research Impact	6402 non-null	float64
8	International Collaboration	6402 non-null	float64
9	Year	6402 non-null	int32
10	Subject	6402 non-null	object
11	Research Impact Factor	6402 non-null	float64
12	Total Score Category	6402 non-null	category
13	Faculty Quality	5700 non-null	category
14	Continent	6402 non-null	object

After

The second dataset was cleaned similarly.

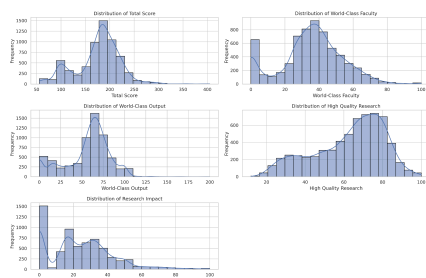
EDA

Key Visuals & Insights

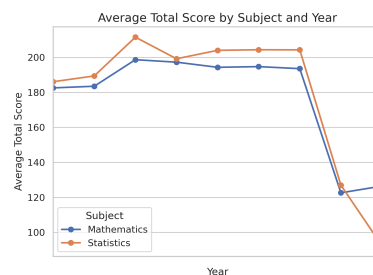
For EDA we gained descriptive statistics via the `df.describe` method on numerical columns.

We also performed:

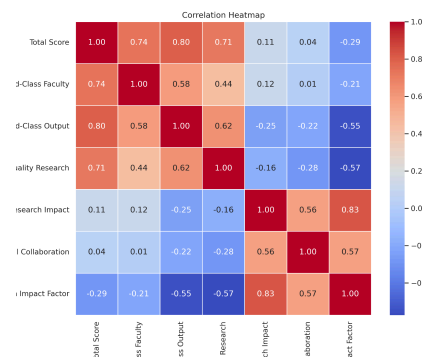
Distribution Analysis



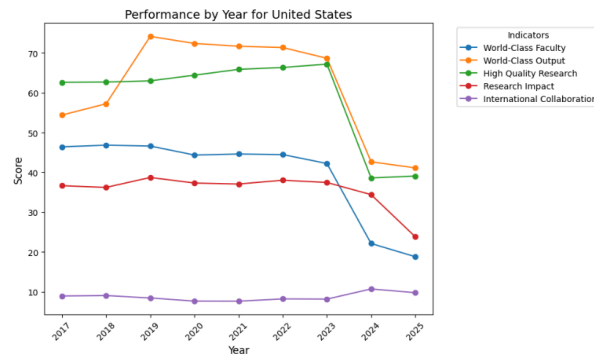
Time-Based Total Score Analysis



Correlation analysis

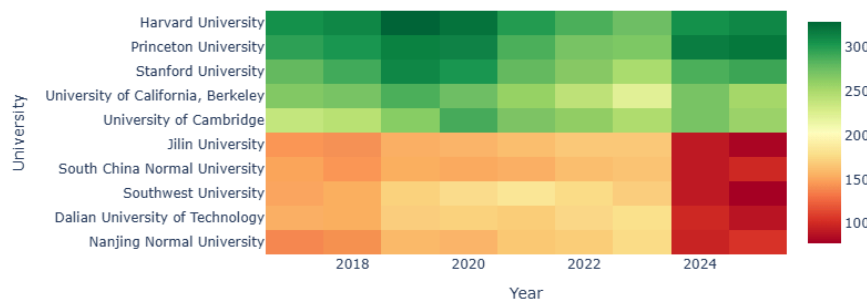


Group by analysis for specific countries



Top-N and Bottom-N records

Top 5 & Bottom 5 Universities (Total Score, 2017–2025)



Notice that top 5 unis are all North American

We Tested 3 Hypothesis:

Hypothesis 1: Regional Performance Disparity

Research Question: Do European countries achieve a higher average total score compared to other continents?

- **Null Hypothesis:** All continents have the same mean total score.
- **Alternative Hypothesis:** Europe has a higher mean total score.
- **Methodology:** One-Way ANOVA followed by Tukey's Honest Significant Difference (HSD) post-hoc test.

Statistical Results

- **ANOVA Result:** The p-value was significantly lower than **0.05**, leading us to **reject the Null Hypothesis**. This confirmed that at least one continent performs differently.
- **Tukey HSD Insight:** While Europe performed strongly, the pairwise comparison revealed that **North America** actually held the highest mean difference in total scores.

Conclusion: We rejected the null hypothesis. However, the data nuanced our original assumption: while regional differences are significant, North America, rather than Europe, leads the average scoring.

Hypothesis 2: Institutional Density vs. Academic Excellence

Research Question: Is there a positive relationship between a country's average score and the number of universities it hosts?

- **Null Hypothesis:** The correlation coefficient is 0.
- **Alternative Hypothesis:** There is a positive correlation.
- **Methodology:** Pearson Correlation Coefficient on aggregated country-level data.

Statistical Results

- **Correlation Coefficient:** 0.2359
- **P-value:** 0.06

Conclusion: Based on the results, we **failed to reject the Null Hypothesis**. There is no statistically significant evidence to suggest that simply having *more* universities leads to a higher *average* score for a country. This suggests that academic excellence is driven by the performance of individual elite institutions rather than the sheer quantity of universities in a region.

Hypothesis 3: Historical Prestige and Performance

Research Question: Do «Old» universities (founded before 1950) perform better than «New» universities?

- **Null Hypothesis:** Total score means of old and new unis are equal.
- **Alternative Hypothesis:** Old unis has greater total score mean.
- **Methodology:** Welch's T-test (independent t-test with unequal variance) after merging the ranking and founding year datasets.

Statistical Results

- **T-statistic:** 3.3692048122562395
- **P-value:** < 0.05

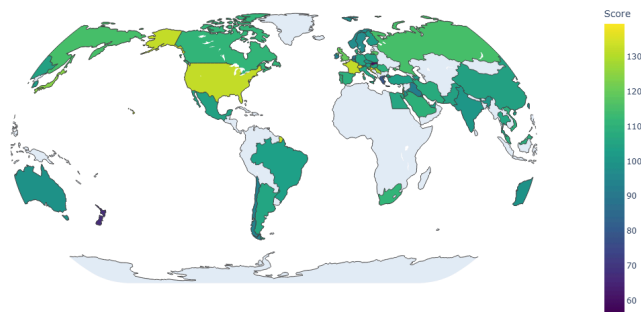
Conclusion: We **rejected the Null Hypothesis**. The statistical evidence strongly supports the alternative hypothesis that older universities (pre-1950) maintain significantly higher total scores than newer institutions. This suggests that factors like long-term reputation, established research infrastructure, and alumni networks contribute to sustained high rankings.

Data Visualization

Key Findings & Insights

1. Coropleth map that shows global university performance.

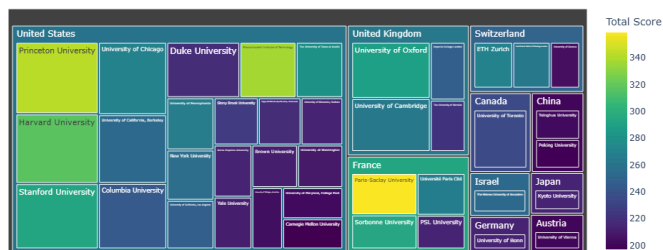
Global Heatmap: Average University Performance (2025)



This interactive choropleth map shows that while high-performing universities are spread across the globe, there is a significant concentration of top-tier average scores in North America, Western Europe, and parts of East Asia.

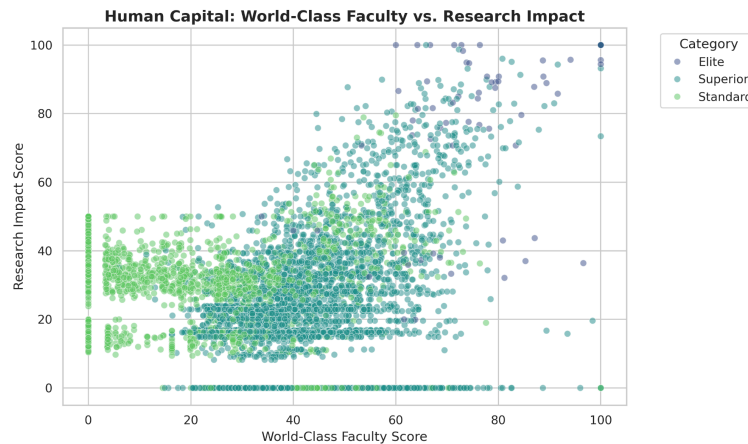
2. This interactive treemap nested by region shows the top 50 universities globally, where the size of each block represents the institution's total score

Top 50 Universities: Hierarchy by Region (2025)



The United States and France dominate the top 50 spots, with a few key regions hosting the majority of the highest-scoring institutions. It visualizes the «concentration of excellence» showing that global academic power is not evenly distributed but clustered respectively to specific economic factors

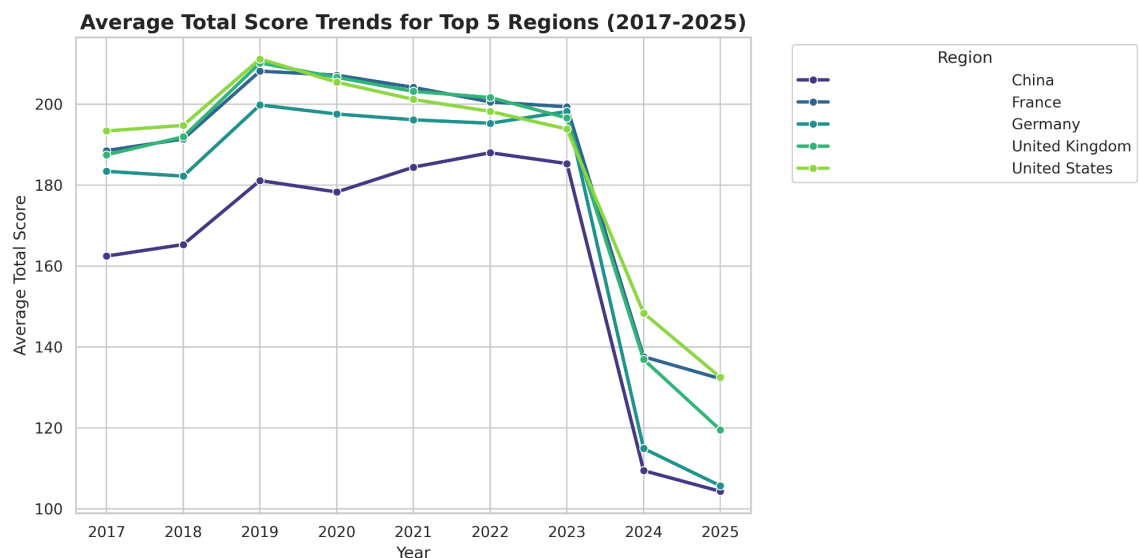
3. The «Human Capital» Hypothesis: Does the presence of «World-Class Faculty» act as a «multiplier» for research impact, or is it just a status symbol?



Most «Standard» universities are clustered at the bottom left, indicating that a certain threshold of faculty quality is often necessary to break into higher tiers of global influence. But also there universities maintaining high Research Impact scores despite having lower World-Class Faculty scores which implies that its not only way to have high impact

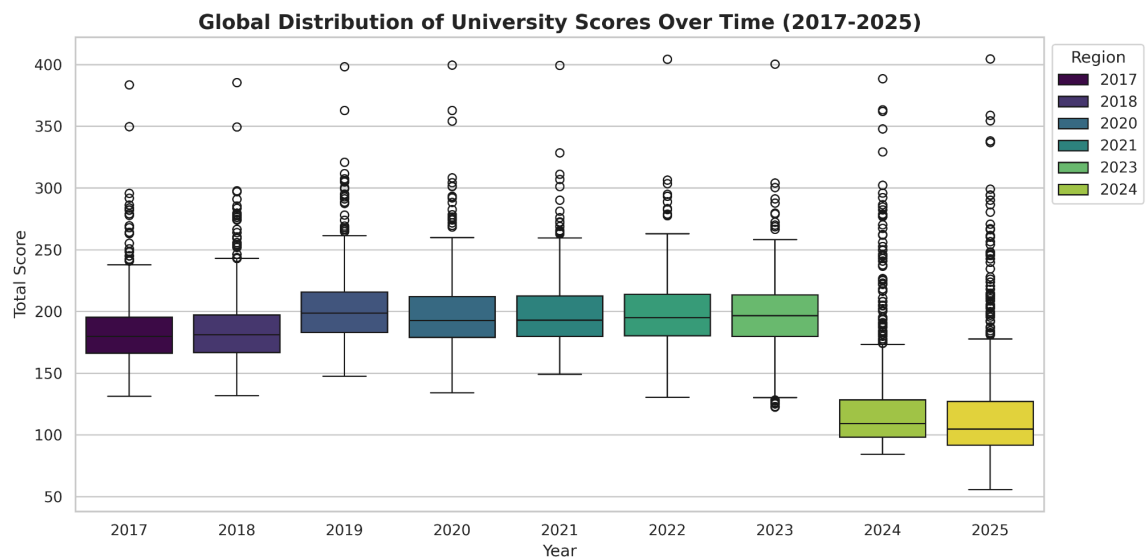
4. Global Distribution of University Scores Over Time (2017-2025)

a. Average Total Score Trends for Top 5 Regions (2017-2025)



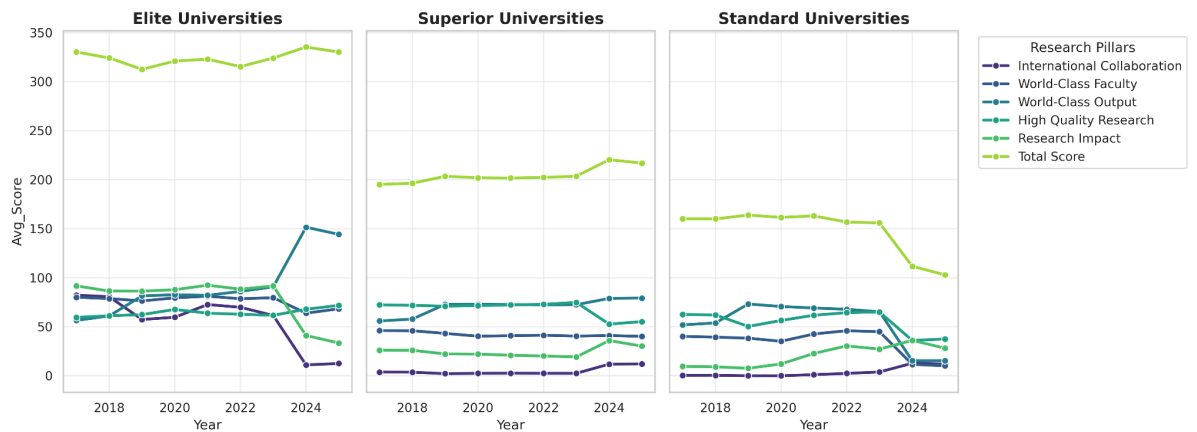
In the first lineplot showed an interesting moment in total scores distribution over the years, its score decline starting in 2023 not only for one country but all over the world and by this we came up with the idea that in this year total score evaluations criteria changed and it is even harder now to get high scores.

b. Global Distribution of University Scores Over Time (2017-2025)



But if we will see next graph which is box plot we can understand that new evaluation structure hasn't really affect elite category universities it is just even more stretched difference between elite and standard universities

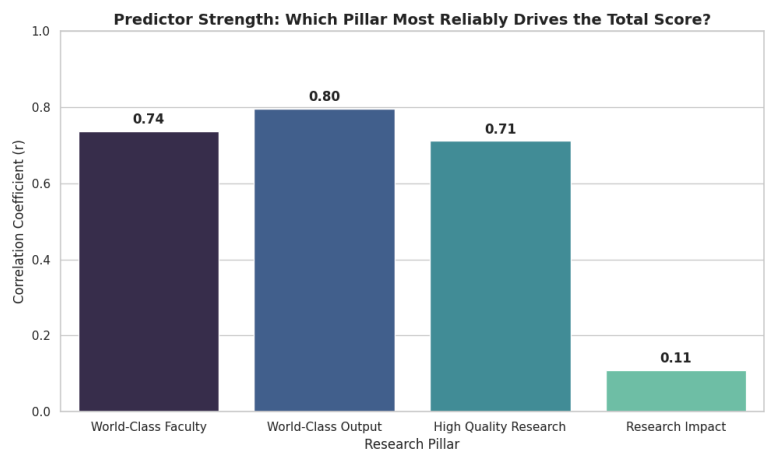
c. The Great Divergence: Pillar Evolution by Institutional Tier (2017-2025)



And then all our conclusions are summed up by the last line plot illustrating distribution for each score types in different categories of universities (elite, superior, standard), there clearly illustrated that only standard universities had declined in 2023 and also in graph it visible reasons why it happened so, this can be explained not because evaluation criteria changed only for standard universities but because elite universities in the same year (2023) had really large outputs score which kept their total score stable and superior universities also boosted their international collaboration, research quality and research impact.

The conclusion is that high-ranked universities have more experience with coping with unexpected changes.

5. The «Quality vs Quantity» Dilemma: Is it better to publish a lot in top journals (World-Class Output) or to publish selectively with extremely high prestige (High Quality Research)?



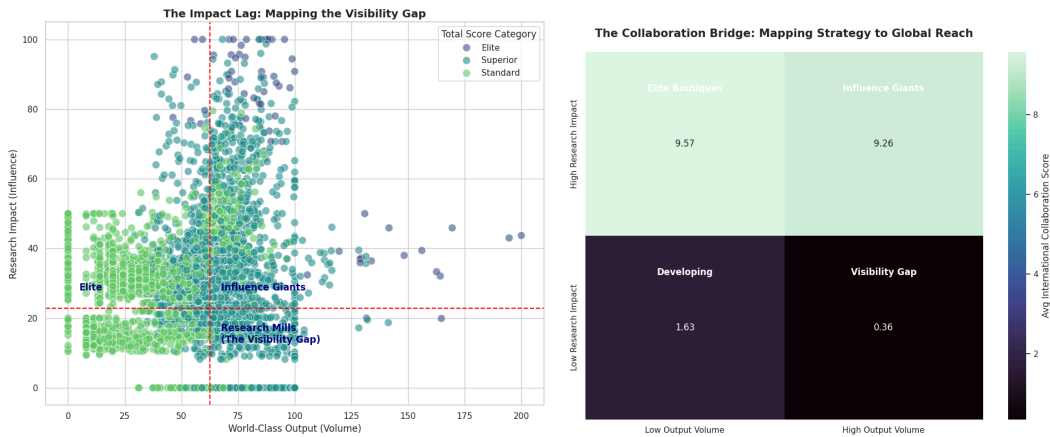
We were suspecting that High Quality Research is a more reliable predictor of Total Score than pure World-Class Output. And here we can see that our hypothesis failed because surprisingly, it tells universities that while prestige is nice, you cannot reach the top without a massive, consistent factory of top-tier outputs.

— Quantity precedes quality

Zhanar

6. Can a university have high World-Class Output but low Research Impact, and vice-versa?

This uncovers the «Visibility Gap» — universities that are doing the work but aren't being cited globally, perhaps due to lack of international collaboration or language barriers.



a. The Impact Lag: Mapping the Visibility Gap

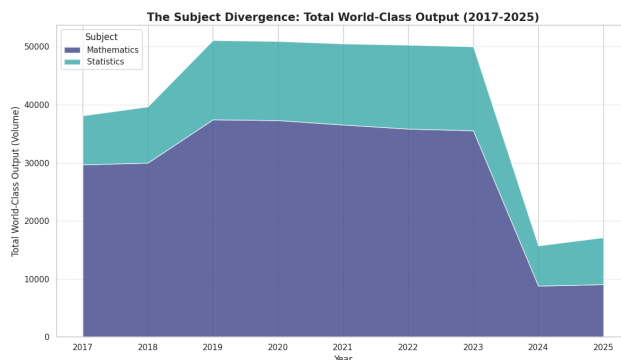
There is a significant segment named «Visibility Gap» of universities that produce high World-Class Output but fail to achieve top-tier Research Impact (High Volume, Low Influence).

b. The Collaboration Bridge: Mapping Strategy to Global Reach

The heatmap's idea is to compute for each type of high/low impact+output its correlation with international collaboration,so this heatmap proves our previous assumption that the visibility gap might be due to lack of global connectivity. Universities that produce high volumes of work but have low impact (the bottom-right cell) have an average collaboration score of 0.36.

For institutional leaders, this heatmap provides a clear directive: *if your department is stuck in the «Visibility Gap» the solution isn't to publish more papers, but to intensify global partnerships.*

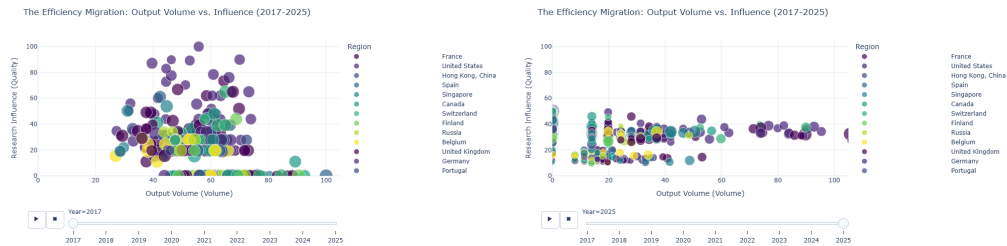
7. The Subject Divergence



Statistics has seen a more aggressive growth in «Volume» over the last decade compared to Mathematics due to the rise of Data Science and AI.

- ❖ Here in this graph shows the ups and downs of subjects over the years by World-Class Output ,so basically the amount of science works made for each subject.
- ❖ So the graph tells that the volume of publications in statistics is significantly higher than in the math field,this might be because universities are increasingly pivoting resources toward Statistics to capture the momentum of the AI era.
- ❖ For academic planners, it shows that the «volume» of prestige is shifting, making Statistics a high-growth sector for talent and funding.

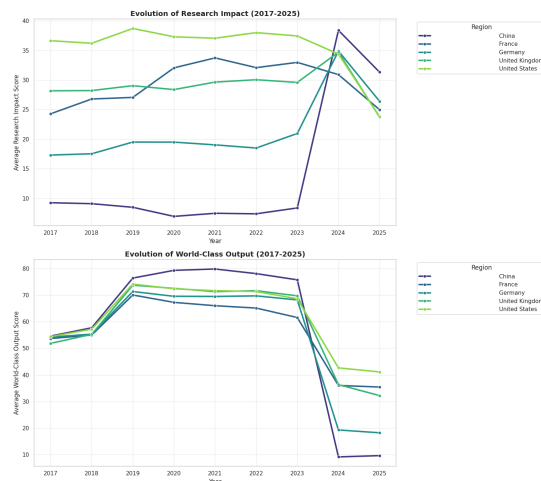
8. Academic history.



This is the ultimate tool for momentum analysis. It proves that institutional rankings are not static prizes but the result of multi-year trajectories. It helps identify which countries have successfully «matured» their academic investments into measurable global influence.

9. Identifying «Rising stars»

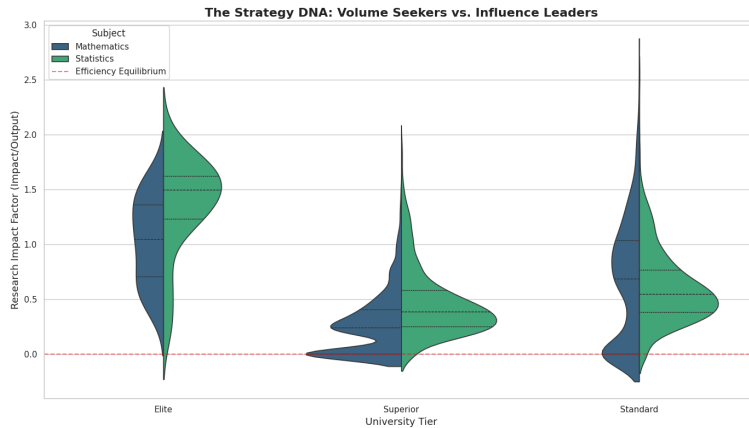
This «Motion Picture» of academic history identifies the future leaders of global research.



- ❖ If a region's Impact is growing faster than its Output, it proves they are becoming more efficient and influential on the world stage.
- ❖ The United States generally maintains high levels in both Research Impact and World-Class Output, indicating consistent performance.
- ❖ China shows a noticeable increase in both metrics, particularly in Research Impact, suggesting a growing efficiency of research.
- ❖ Other regions like France, Germany, and the United Kingdom exhibit varying trends, with some years showing fluctuations or more modest growth compared to the larger players.

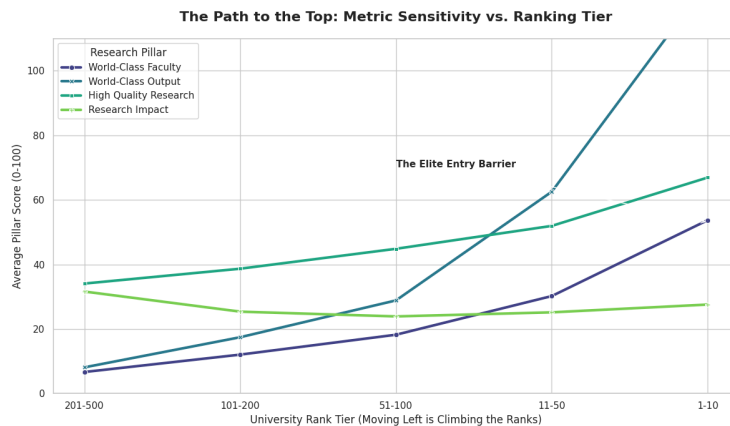
10. Are Elite universities at the top because they produce more work or because their work is higher quality relative to its volume?

This violin plot illustrates how rank and publishing quantity and quality are aligned, and distributed along the tree category .



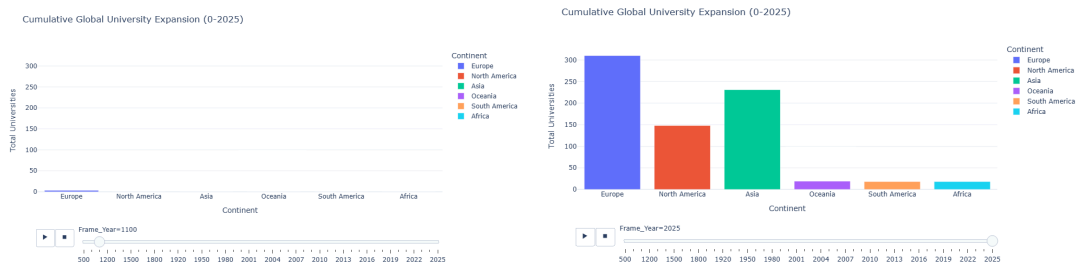
We can see the «Elite» violin bulges significantly above the 1.0 Equilibrium line, it proves that these institutions don't just publish a lot; they publish better. Their quality score outpaces their output volume, identifying them as «Prestige Leaders». Also interesting that mathematics has a lower impact factor which tells us its relatively harder to make an impact in the math field with average publications.

11. The «Path to the Top» (Metric Sensitivity)



You will notice that World-Class Output and Research Impact often show the steepest growth in these tiers. This confirms that for universities moving into the Top 100, increasing volume and visibility is the primary driver.

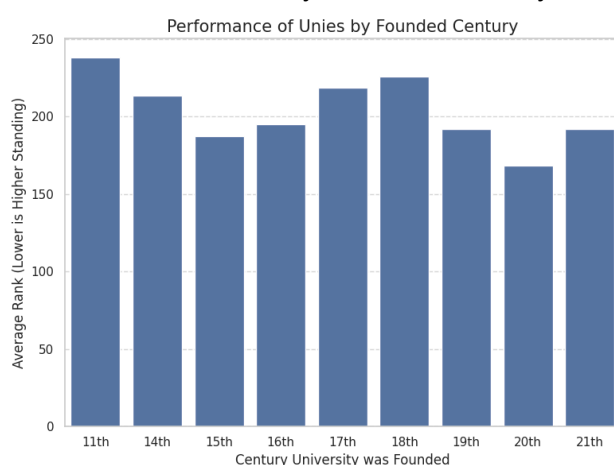
12. Historical University distribution



If you watch the early frames of the animation (especially for years before, say, the 17th or 18th century), you'll notice that the bars for Europe will appear first and grow significantly, indicating that the oldest universities and the initial major expansion of higher education were indeed concentrated in European countries.

Universities in Every Continent by Early 20th Century: As the year progresses towards the early 1900s (around 1900-1920), bars representing universities begin to appear and grow for virtually all continents (North America, Asia, South America, Africa, Oceania). This confirms that by the early 20th century, the establishment of universities had become a global phenomenon, with institutions founded in every major continent.

13. Performance of Unies by Founded Century



Higher Scores for Older Institutions: Generally, universities founded in earlier centuries (e.g. 14th-19th centuries) tend to have higher average total scores compared to those established more recently.

Peaks of Excellence: There are noticeable peaks in average performance for universities founded in the 14th century, and again for institutions from the 19th and early 20th centuries, suggesting periods of significant academic establishment and growth.

Recent Trends: Universities founded in the 21st century currently show a lower average total score. This could be attributed to these institutions still being relatively young and in the process of building their reputation and research output, or it might also be influenced by the recent 2023-2024 Pivot in ranking methodologies that we discussed earlier, potentially affecting newer institutions differently.

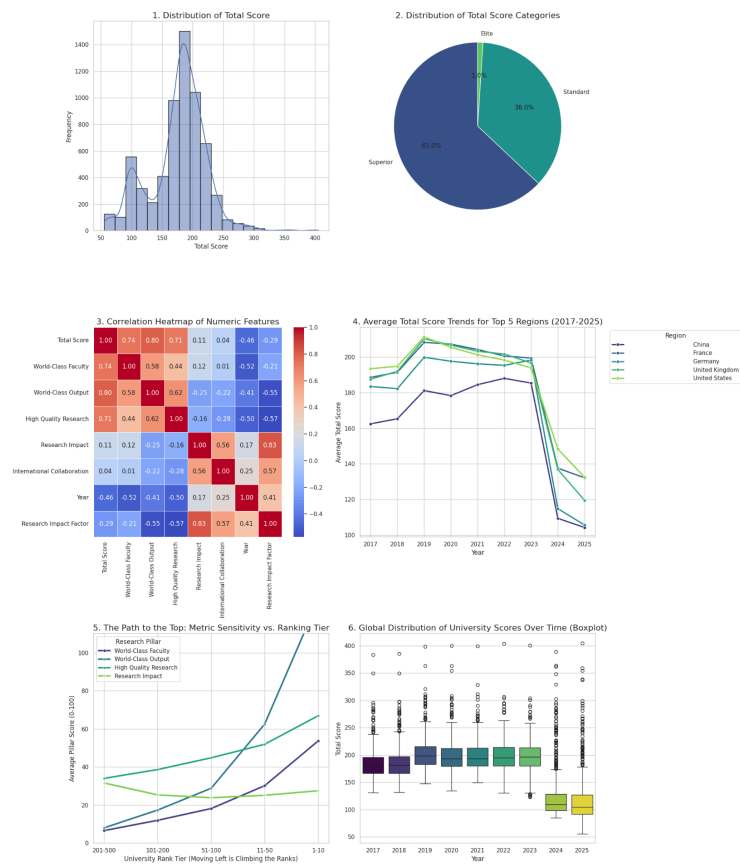
Elite Performance: Universities categorized as «Elite» tend to demonstrate high scores in both «World-Class Faculty» and «Research Impact», clustering towards the top-right corner of the plot. This suggests that a strong faculty presence is often coupled with significant research influence for top-tier institutions.

Minimum Threshold for Impact: Most «Standard» universities are concentrated in the lower-left quadrant, indicating that a certain level of «World-Class Faculty» score is generally necessary to achieve higher «Research Impact». It appears challenging to

achieve high research influence without a foundational level of faculty excellence.

Beyond Faculty: While a high «World-Class Faculty» score is common among «Elite» institutions, the graph also shows instances where universities maintain high 'Research Impact' scores despite having comparatively lower «World-Class Faculty» scores. This implies that while faculty quality is crucial, it is not the sole determinant of research influence; other factors might contribute significantly to a university's impact.

14. Summary Dashboard



The dashboard paints a picture of a highly competitive and stratified global academic environment. Success in rankings is driven by robust research ecosystems, strong faculty, and impactful output. The recent changes in ranking methodologies (the «2023-2024 pivot») have introduced new challenges, potentially exacerbating the existing disparities between elite and other institutions, and emphasizing the need for strategic adaptation to maintain or improve global standing.

Machine Learning

Project Objective: Comparative Analysis of Academic Disciplines using Machine Learning

Goal: To determine whether «Mathematics» and «Statistics» can be distinguished based on their academic performance metrics and to identify the key features that define each subject's unique profile.

Data: Using university ranking metrics (Research Impact, Faculty quality, etc.).

- **Model:** Random Forest Classifier
- **Evaluation:** Confusion Matrix

Conclusions After Model Execution:

- **High Model Accuracy:** The Random Forest model successfully distinguished between the two subjects with high precision. Looking at the Confusion Matrix, the model correctly identified over 90% of cases, suggesting that Mathematics and Statistics have distinct "academic signatures."
- **Key Differentiator: Research Impact** is the most significant factor (Importance: 0.33) in separating these disciplines. This indicates that the way research influence is distributed varies greatly between Mathematics and Statistics departments globally.
- **Academic Profiles:**
 - **Mathematics:** Displays a more "traditional" profile, remaining stable even with lower "World-Class Faculty" scores.
 - **Statistics:** Shows a stronger reliance on "International Collaboration" and "World-Class Output" for its high-ranking positions.
- **Final Insight:** While Statistics is often seen as a branch of Mathematics, this ML analysis proves they are distinct entities in the global ranking ecosystem, driven by different performance drivers.

Challenges Faced

- ❖ We faced a lot of challenges during the scraping stage due to the dynamic nature of the website. Several strategies had been tested to cope with the second table, first we tried to convert each university name into a link, but it appeared that some links did not exist or were written differently, so eventually we decided to stick to the selenium.
- ❖ The second table caused a plethora of troubles during scraping, since some buttons weren't clickable. We solved this problem by implementing a recovery mechanism.
- ❖ During cleaning, we noticed that some columns in the second table are almost fully null. We dropped them, since no other worthy alternative came to our minds.

Individual Contributions of Each Team Member

Stage	Owner
Scraping	Zhanar
Preprocessing	Akmeiir

EDA	Fazilat
Hypothesis Testing	Fazilat
Visualization	Feruza
Visualization Analysis	Feruza
Machine Learning	Akmeiir
Report & Presentation	Zhanar